Life after disturbance

Secondary succession occurs after a disturbance that destroys plant life, such as a flood or a fire. The plant life reestablishes itself and develops into an ecosystem similar to the one that existed before the disturbance. The stages of secondary succession are similar to those of primary succession, although the ecosystem may start at different points in the process, depending on the level of damage caused by the trigger.

A common example of secondary succession occurs after a wildfire in oak and hickory forests. Nutrients from burned plants and animals provide the right conditions for growth of annual plants. Pioneer grasses soon follow. After several years, due at least in part to the environmental and soil changes resulting from the pioneer species, shrubs and oak, pine, and hickory trees will begin to grow. As the trees grow higher, shading out more of the underbrush, the grasses are replaced by plants able to survive with low sunlight, and, after around 150 years, the forest once more resembles the prefire community.

"I … found indisputable evidence (a) that forests succeeded prairie, and (b) that prairie had succeeded forest."

Henry Allan Gleason

American ecologist

See also: Field experiments • The ecosystem • Climax community • Open community theory • Biomes • Romanticism, conservation, and ecology